

NORTHWAY, AK, USA

WMO: 702910

Lat: 62.961N

Lon: 141.929W

Elev: 1713

StdP: 13.81

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26412

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.9	-43.6	-34.6	0.8	-28.2	-33.0	0.9	-27.3	15.4	-3.0	12.2	0.0	0.1	80	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	78.3	58.6	74.8	57.2	71.7	56.0	60.7	74.4	59.0	71.4	57.3	68.3	5.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.4	69.7	61.7	53.9	65.9	60.8	52.3	62.0	59.9	27.6	74.4	26.4	71.6	25.3	68.5	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.1	10.6	DB	-50.4	83.4	6.1	3.2	-54.8	85.7	-58.3	87.6	-61.7	89.4	-66.2	91.7	(4)
(5)			WB	-34.9	64.9	3.8	4.7	-37.6	68.3	-39.9	71.0	-42.0	73.7	-44.8	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	-13.6	-3.8	7.4	30.1	46.4	56.5	59.2	54.2	43.0	-0.9	-11.2	(6)	
(7)		DBStd	28.63	15.96	13.91	13.52	10.00	7.36	5.59	4.48	5.65	6.86	10.58	13.80	13.02	(7)
(8)		HDD50	10069	1973	1507	1321	596	159	8	2	22	219	837	1528	1897	(8)
(9)		HDD65	14860	2438	1927	1786	1046	576	263	186	337	659	1302	1978	2362	(9)
(10)		CDD50	700	0	0	0	0	49	202	286	153	10	0	0	0	(10)
(11)		CDD65	15	0	0	0	0	0	7	6	2	0	0	0	0	(11)
(12)		CDH74	375	0	0	0	0	20	138	145	72	0	0	0	0	(12)
(13)		CDH80	52	0	0	0	0	1	28	16	7	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	1.2	1.8	3.0	4.1	5.1	4.7	4.3	3.8	3.6	2.9	1.6	1.2	(14)
(15)	Precipitation	PrecAvg	9.6	0.3	0.3	0.2	0.2	0.9	1.9	2.4	1.5	1.1	0.5	0.3	0.3	(15)
(16)		PrecMax	15.2	1.0	1.2	0.6	0.7	2.6	4.4	5.0	3.0	3.5	1.3	1.1	0.8	(16)
(17)		PrecMin	5.7	0.0	0.0	0.0	0.0	0.0	0.4	0.3	0.0	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	2.2	0.2	0.3	0.2	0.2	0.7	0.9	1.2	0.7	0.7	0.3	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	29.0	45.4	62.9	76.8	84.5	82.3	80.9	66.5	49.9	28.8	21.8	(19)
MCWB			25.5	26.2	36.4	46.8	53.8	61.6	60.3	60.7	51.0	40.7	27.6	21.0	(20)	
2%		DB	17.0	24.1	39.2	54.4	71.6	77.6	78.1	75.3	62.7	43.4	24.4	13.8	(21)	
		MCWB	16.3	22.1	32.0	41.6	52.8	57.7	59.2	59.0	49.4	37.4	23.6	12.8	(22)	
5%		DB	11.5	19.3	34.2	50.6	67.0	73.7	74.6	70.8	59.4	38.8	20.4	10.1	(23)	
		MCWB	10.7	17.9	28.7	39.1	49.6	55.9	58.0	56.6	47.9	34.8	19.8	9.2	(24)	
10%		DB	7.1	15.3	29.5	47.3	62.8	70.1	71.5	66.8	56.3	35.4	17.1	6.9	(25)	
		MCWB	6.2	14.2	25.3	37.3	47.7	54.7	56.9	54.5	46.6	33.0	16.4	6.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.5	37.2	47.4	56.2	63.9	62.1	62.4	53.1	42.0	27.8	21.7	(27)
MCD B			27.4	29.1	44.5	61.1	71.5	77.5	76.4	76.4	62.6	47.1	28.8	22.8	(28)	
2%		WB	17.3	22.4	32.3	42.4	53.1	60.3	60.5	59.8	50.9	37.8	23.7	13.8	(29)	
		MCD B	18.0	24.4	38.5	53.5	69.7	74.8	74.0	72.4	59.8	42.6	24.5	14.5	(30)	
5%		WB	11.8	18.5	29.0	39.6	50.8	57.6	59.2	57.7	49.3	35.2	19.8	9.9	(31)	
		MCD B	12.5	19.7	34.3	49.1	64.6	70.1	71.4	68.2	57.4	38.2	20.4	10.6	(32)	
10%		WB	7.7	14.6	25.3	37.6	48.7	55.5	57.8	55.6	47.5	33.0	16.6	6.7	(33)	
		MCD B	8.6	15.7	29.6	46.1	60.9	67.3	68.9	64.8	55.0	35.1	17.3	7.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.0	20.9	29.9	25.5	23.2	21.7	20.2	20.2	20.0	15.1	14.2	14.4	(35)
MCDBR			16.2	21.5	31.7	27.9	30.8	28.1	27.1	27.7	26.2	19.2	12.6	15.9	(36)	
MCWBR			15.8	19.9	24.6	16.4	13.7	11.6	11.1	12.7	14.2	13.5	12.2	15.7	(37)	
5% WB		MCD BR	15.9	21.1	29.8	26.3	28.0	24.6	23.0	24.4	22.6	17.1	12.7	16.1	(38)	
		MCWBR	15.6	19.5	23.1	15.7	12.8	11.5	10.2	11.8	13.0	12.3	12.3	15.9	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub		0.201	0.242	0.259	0.333	0.349	0.384	0.377	0.372	0.299	0.267	0.187	0.170	(40)
taud		1.997	2.247	2.271	2.172	2.391	2.307	2.357	2.378	2.577	2.477	2.058	1.924	(41)		
Ebn at Noon		142	232	272	267	275	265	264	254	260	224	152	86	(42)		
Edh at Noon		10	18	26	37	33	36	34	30	20	15	9	5	(43)		
(44)	All-Sky Solar Radiation	RadAvg		72	309	862	1447	1676	1728	1559	1253	790	348	103	32	(44)
(45)		RadStd		6	24	59	87	76	119	106	104	62	39	8	2	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.62	+4.03	N/A	N/A	N/A	N/A	-536	-608	N/A	N/A
(47)	Regional (1 neighbor)	+1.39	+4.32	+2.32	N/A	N/A	N/A	-477	-508	+92	N/A

Nomenclature: See separate page